Electrical Design and Layout of a 110 kWp Rooftop Grid-Connected Photovoltaic System

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Abstract

This work documents the electrical design of a 110 kWp grid-connected photovoltaic (PV) array placed on the flat roof of an industrial building. The design fixes the number of modules per string from the low-temperature open-circuit voltage limit of the inverter, checks the maximum power point tracking (MPPT) window at high cell temperature, and sizes the inter-row spacing from the winter-solstice shading geometry at latitude 37°. Twenty modules per string keep the cold-string voltage at 1084 V, safely below the 1100 V inverter input ceiling, and ten strings give a DC/AC ratio of 1.10. The results are expressed as two CAD sheets—a site layout and a low-voltage single-line diagram—generated programmatically so that every dimension traces back to the stated parameters.

1 Introduction

Rooftop PV plants in the 50–250 kWp range are a common way for industrial consumers to offset daytime load. For a string inverter the first design decision is the string length, because the array voltage must never exceed the inverter input rating, yet must stay inside the MPPT window across the full operating temperature span (Masters , 2013). The open-circuit voltage of a crystalline-silicon module rises as temperature falls, so the binding case is the coldest expected cell temperature rather than standard test conditions (Luque & Hegedus , 2011). Alongside the voltage chain, the layout must leave enough spacing between tilted rows to avoid mutual shading near solar noon in winter. This paper carries a single 550 Wp module and a 100 kW inverter through these checks and renders the outcome as dimensioned drawings. The requirements of IEC 62548 for PV array design (International Electrotechnical Commission , 2016) and IEC 60364-7-712 for PV installations (International Electrotechnical Commission , 2017) frame the protection and earthing choices.

2 Method

2.1 Components

The array uses a 550 Wp monocrystalline module ($V_{oc} = 49.5$ V, $V_{mp} = 41.7$ V, $I_{sc} = 13.95$ A, $I_{mp} = 13.19$ A, $\beta_{Voc} = -0.27$ %/°C, 2278×1134 mm). The inverter is a 100 kW unit with an MPPT window of 200–1000 V and a maximum DC input of 1100 V. Table 1 collects the figures used below.

2.2 String length from the cold-voltage limit

With the coldest design cell temperature taken as −10°C, the per-module open-circuit voltage is

$$V_{oc}(T) = V_{oc,\mathrm{STC}}\,[1 + \beta_{Voc}\,(T - 25)] = 49.5\,[1 + (-0.0027)(-10 - 25)] = 54.18 \text{ V}. \tag{1}$$

Table 1: Module and inverter parameters used in the design.

Item	Symbol	Value
Module power (STC)	P_{mp}	550 Wp
Open-circuit voltage	V_{oc}	49.5 V
MPP voltage	V_{mp}	41.7 V
V_{oc} temp. coefficient	β_{Voc}	-0.27 %/°C
Inverter max DC input	$V_{dc,max}$	1100 V
Inverter MPPT window	–	200–1000 V

The number of modules per string follows from the inverter ceiling:

$$N_{max} = \frac{V_{dc,max}}{V_{oc}(-10°\mathrm{C})} = \frac{1100}{54.18} = 20.3 \;\Rightarrow\; N = 20, \tag{2}$$

giving a worst-case string voltage of $20 \times 54.18 = 1083.6$ V, below the 1100 V limit.

2.3 MPPT window at high temperature

Taking $\beta_{Vmp} \approx -0.34$ %/°C and a hot cell at 70°C,

$$V_{mp}(70°\mathrm{C}) = 41.7\,[1 + (-0.0034)(70 - 25)] = 35.3 \text{ V}, \tag{3}$$

so the string works at $20 \times 35.3 = 706$ V when hot and 834 V at STC—both inside the 200–1000 V window.

2.4 Row spacing from shading geometry

For a module tilt of 15° the height and base projection of a landscape module (1.134 m along the slope) are $h = 1.134 \sin 15° = 0.293$ m and $b = 1.134 \cos 15° = 1.095$ m. The winter solstice noon solar altitude at latitude 37° is $\alpha = 90° - 37° - 23.45° = 29.55°$, so the shadow reaches $d = h/\tan\alpha = 0.517$ m and the minimum pitch is $p_{min} = b + d = 1.61$ m. A pitch of 3.0 m was adopted to add a maintenance walkway.

3 Results

The array is organised as ten strings of twenty modules (200 modules, 110.0 kWp), which against the 100 kW inverter gives a DC/AC ratio of 1.10. Figure 1 shows the generated site plan: the ten colour-coded strings, the DC collection bus feeding the inverter and the AG distribution panel, the perimeter earthing ring with corner electrodes, and the dimensioned roof outline with the title block.

Figure 2 gives the low-voltage single-line diagram. On the DC side each string carries a 16 A gPV fuse, a Type II DC surge arrester and an 1100 V DC disconnector; on the AC side the inverter feeds a 160 A moulded-case breaker, a bidirectional meter and the AG distribution panel before the grid. The 160 A breaker follows from the AC current $I_{ac} = 100\,000/(\sqrt{3} \cdot 400 \cdot 0.98) \approx 147$ A.

Table 2 summarises the sizing outcome. Every value in the drawings is derived from the parameters in Table 1; changing a parameter in the generating script updates the sheets consistently.

4 Conclusion

A compact, parameter-driven procedure fixed the string length from the cold open-circuit voltage, confirmed the MPPT window at high temperature, and set the row pitch from winter shading,

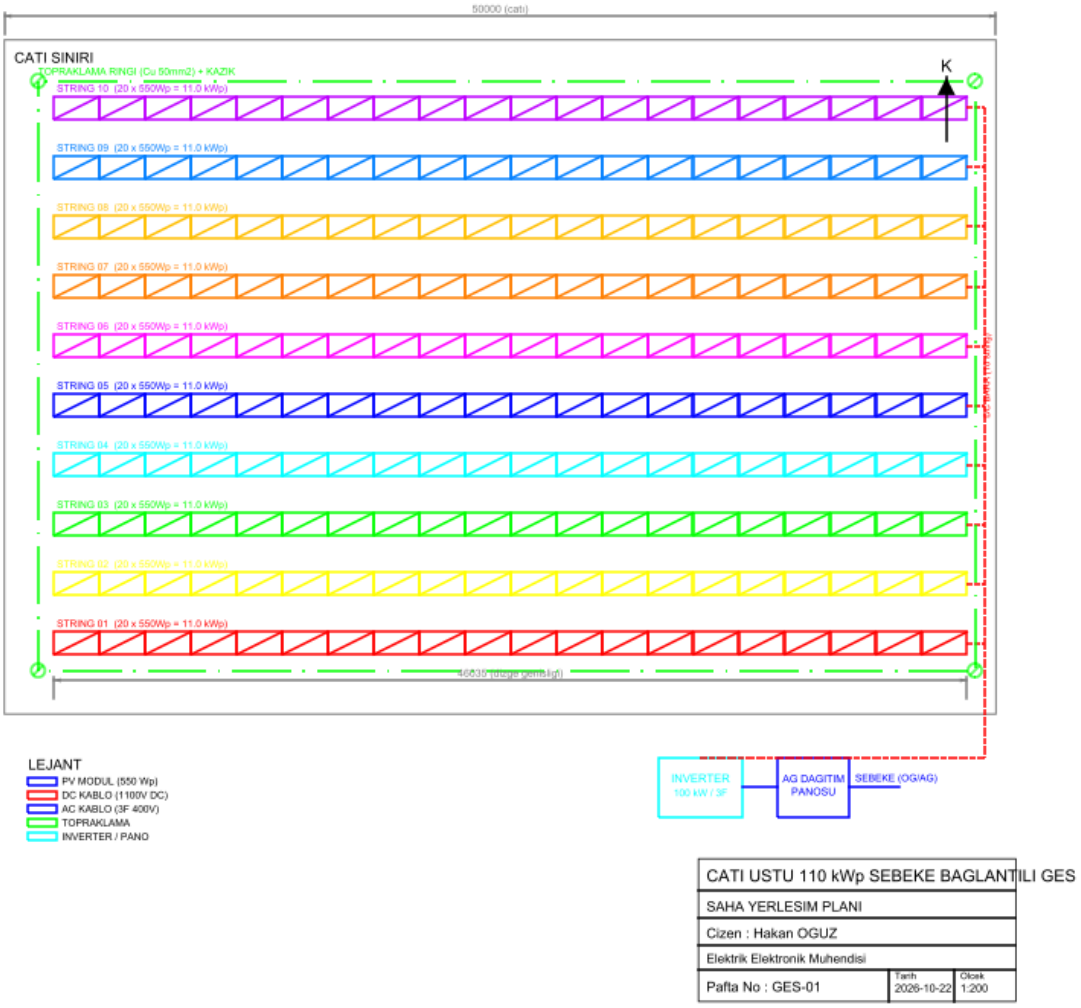


Figure 1: Generated site layout (sheet GES-01): ten strings, DC bus, inverter/panel, earthing ring.

yielding a 110 kWp array of ten twenty-module strings on a 100 kW inverter. Because the two CAD sheets are produced from the same parameter set, the layout and the single-line diagram stay internally consistent and are easy to revise for a different module or roof. The drawings are at preliminary design level; a construction package would add structural, detailed cable-drop and utility interconnection work.

References

International Electrotechnical Commission. (2016). *IEC 62548:2016 Photovoltaic (PV) arrays – Design requirements*. IEC.

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Luque, A., & Hegedus, S. (Eds.). (2011). *Handbook of photovoltaic science and engineering* (2nd ed.). Wiley.

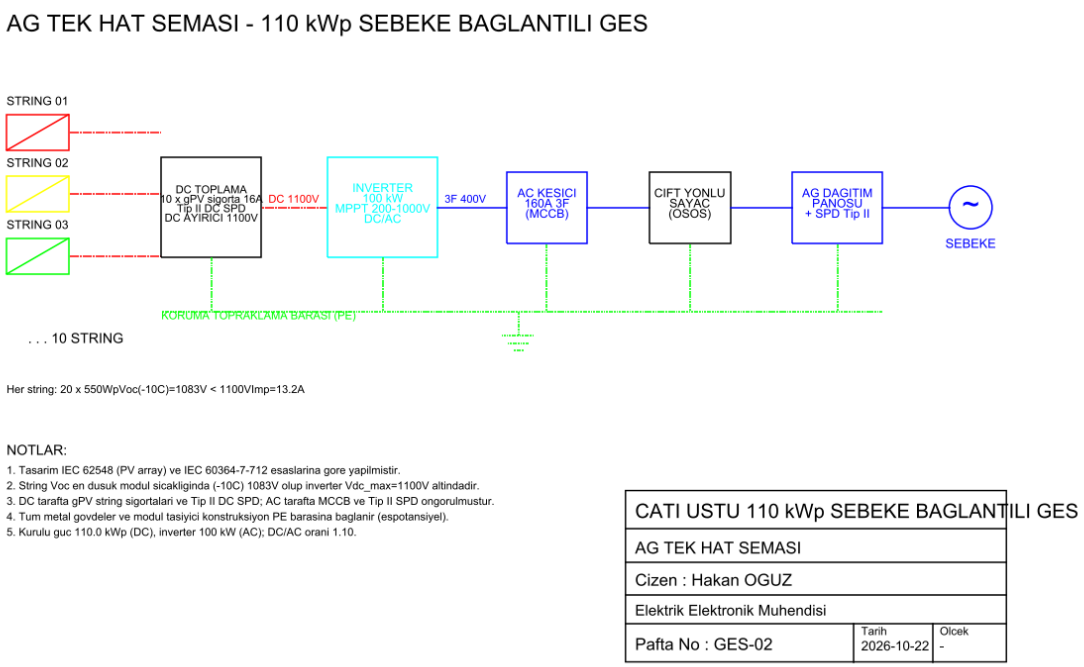


Figure 2: Generated single-line diagram (sheet GES-02) from string to grid connection.

Table 2: Design outcome.

Quantity	Value
Modules per string	20
Number of strings	10
Installed DC power	110.0 kWp
Cold string voltage (-10°C)	1083.6 V
Hot string MPP voltage (70°C)	706 V
DC/AC ratio	1.10
AC current / main breaker	147 A / 160 A
Adopted row pitch	3.0 m

Masters, G. M. (2013). *Renewable and efficient electric power systems* (2nd ed.). Wiley.